Project Design Phase

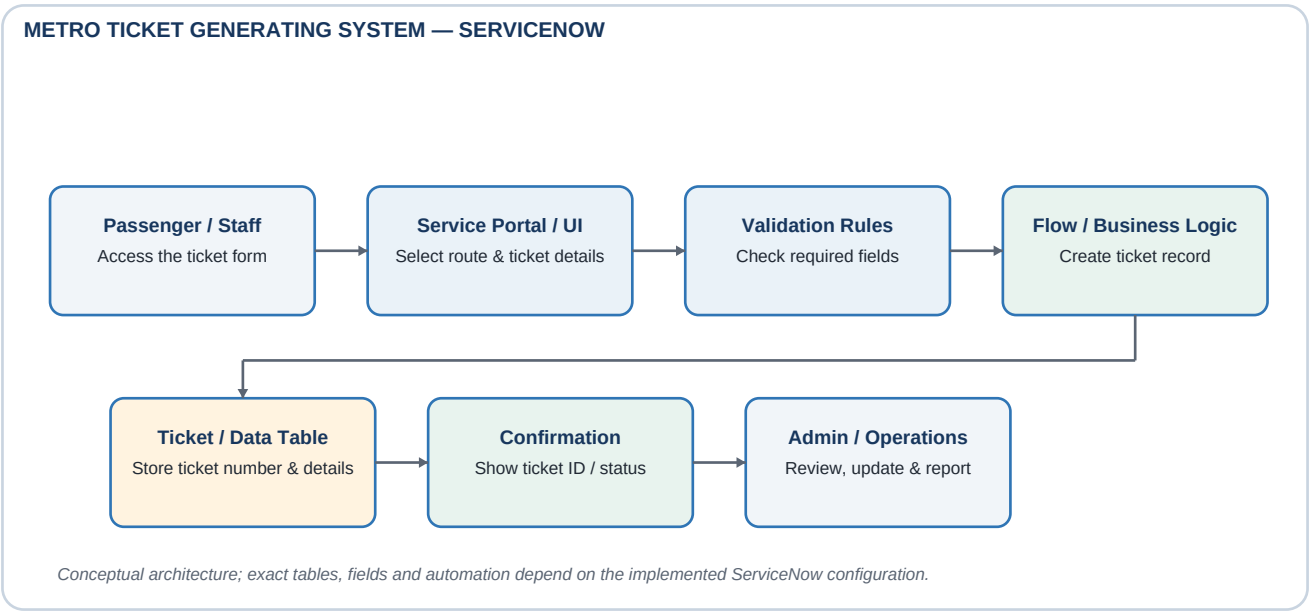
# Solution Architecture

Metro Ticket Generating System in ServiceNow

| Date | 16 October 2026 | Maximum Marks | 4 Marks |
|---|---|---|---|

## Solution Architecture

Solution architecture describes how the Metro Ticket Generating System connects its users, ticket-entry interface, validation and automation logic, stored ticket records, and confirmation process. The design below is a conceptual view for a ServiceNow implementation; exact configuration details should follow the tables, fields, roles, and flows built in the application.




## Main Components and Responsibilities

| Component | Responsibility |
|---|---|
| Passenger / Staff | Enters ticket details such as route or station, journey information, and other required inputs. |
| Service Portal / User Interface | Presents the ticket form and guides the user through submitting a ticket request. |
| Validation Rules | Checks mandatory fields and validates submitted information before a record is created. |
| Flow / Business Logic | Processes a valid submission and creates or updates the relevant ServiceNow ticket record. |
| Ticket Data Table | Stores the generated ticket identifier, ticket details, status, and related record information. |
| Confirmation and Admin | Returns the ticket reference/status to the user and supports authorized staff in reviewing, updating, and reporting on records. |

## Typical Data Flow

**1.** The user opens the ticket form and enters the required journey/ticket information.

**2.** The application validates the submitted fields and displays errors if required information is missing or invalid.

**3.** If validation succeeds, ServiceNow automation creates a ticket record and assigns a unique reference according to the configured implementation.

**4.** The system displays a confirmation and ticket reference; authorized staff can review or update the record.

## Architecture Notes

This architecture is based on the project scope and requirements available so far. External payment gateways, live metro APIs, QR-code scanning, and third-party integrations are not assumed here because they were not specified in the supplied project requirements. They can be added only if included in the final design.